AP CALCULUS AB

Unit 1 • Unit 1 Pretest • Limits, Continuity, and Differentiability • Answer key

Unit 1 Pretest • Limits, Continuity, and Differentiability

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Part A • Multiple choice

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$ is

(A) 0

(B) 3

(C) 6

(D) nonexistent

Solution

Answer: (C).

Question 2

NO CALCULATOR

$\lim_{x \to \infty} \dfrac{5x^2 - x}{2x^2 + 3}$ is

(A) 0

(B) $\frac{5}{2}$

(C) 5

(D) ∞

Solution

Answer: (B).

Question 3

NO CALCULATOR

$\lim_{x \to 1^-} \dfrac{2}{x-1}$ is

(A) ∞
(B) 2
(C) 0
(D) $-\infty$

Solution
Answer: (D).

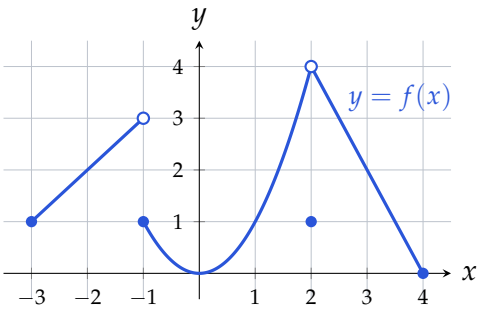
Question 4

NO CALCULATOR

$\lim_{x \to 2} \dfrac{x^2+x-6}{x-2}$ is

(A) 0
(B) 1
(C) 5
(D) nonexistent

Solution
Answer: (C). $\frac{(x-2)(x+3)}{x-2} = x+3 \to 5$.



Question 5

NO CALCULATOR

Using the graph of f, which limit does not exist?

(A) $\lim_{x \to 0} f(x)$
(B) $\lim_{x \to -1} f(x)$
(C) $\lim_{x \to 2} f(x)$
(D) $\lim_{x \to 3} f(x)$

Solution
Answer: (B).

Question 6

NO CALCULATOR

Using the graph of f, at which x is the discontinuity removable?

(A) $x = -1$
(B) $x = 0$
(C) $x = 2$
(D) $x = 3$

Solution
Answer: (C).

Question 7

NO CALCULATOR

For which k is $f(x) = \begin{cases} kx^2, & x \leq 1 \\ x+3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

Solution
Answer: (A).

Question 8

NO CALCULATOR

f is continuous on $[2,6]$, $f(2) = 5$, $f(6) = -3$. Which must be true?

(A) $f(4) = 1$
(B) $f(c) = 0$ for some c
(C) f has no maximum
(D) $f'(c) = -2$

Solution
Answer: (B). IVT. (D) needs differentiability.

Question 9

NO CALCULATOR

f is differentiable, $f(1) = 2$ and $f(5) = 10$. Which must be true?

(A) $f'(c) = 2$ for some c in $(1,5)$
(B) $f'(c) = 8$ for some c in $(1,5)$
(C) $f(3) = 6$
(D) $f'(c) = 0$ for some c in $(1,5)$

Solution
Answer: (A). MVT: $\frac{10-2}{5-1} = 2$.

Question 10

NO CALCULATOR

$f(x) = |x+1|$ is not differentiable at

(A) $x = 1$
(B) $x = 0$
(C) $x = -1$
(D) nowhere

Solution
Answer: (C).

Part B • Free response

Question 11 (Free response)

NO CALCULATOR

Let $g(x) = \begin{cases} \dfrac{x^2+x-6}{x-2}, & x < 2 \\ cx+1, & x \geq 2. \end{cases}$

(a) Find $\lim_{x \to 2^-} g(x)$.
(b) Find c so that g is continuous at $x = 2$; justify with the definition.
(c) With this c, is g differentiable at $x = 2$? (The left piece simplifies to $x + 3$.)

Solution
(a) $x + 3 \to 5$. (b) $2c + 1 = 5$, $c = 2$; $g(2) = 5 =$ lim. (c) Left slope 1, right slope 2: not differentiable (corner).

Question 12 (Free response)

NO CALCULATOR

A differentiable function W (temperature, °C) has $W(0) = 20$, $W(3) = 24$, $W(5) = 31$, $W(8) = 27$.

(a) Must $W(t) = 30$ for some t in $(3, 5)$? Justify.
(b) Must $W'(t) = 3.5$ for some t in $(3, 5)$? Justify.
(c) Must W have an absolute maximum on $[0, 8]$? Justify.

Solution
(a) Yes, IVT ($24 < 30 < 31$; differentiable implies continuous). (b) Yes, MVT ($\frac{31-24}{2} = 3.5$). (c) Yes, EVT: W is continuous on the closed interval $[0, 8]$; the maximum is at least 31.